4.7 Applications

I'd like to let you think about this consideration: it is pretty likely that some software you are using right now is leveraging Bloom filters. In fact, if you are reading the digital version of this book, then it's 100% sure that its download leveraged Bloom filters, as it's common for Internet nodes to use them for their routing tables[13].

4.7.1 Cache

With *caching* we refer to the process of storing some information retrieved on a fast storage system A, in order to have it ready in case we need to read it again in the close future. The data might have been previously retrieved from a slow(er) storage system B, or be the result of a CPU-intensive computation.

For web applications scalability is a major concern, probably the one aspect that is most debated in design reviews for any product that aspires to become viral.

One of my favorite books tackling design and scalability, Scalability Rules, has a chapter titled "Use Caching Aggressively", where it is shown how caching at all levels of a web app is fundamental to allow its scaling.

Cache is often the only thing saving databases from literally ging on fire - or at least crash. But even your laptop has several levels of caching, from fast L1-cache inside its CPU to in-memory cache used to process big files. Your very *operating system* caches memory pages in and out of RAM,